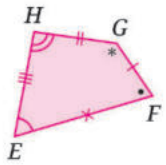
Congruence of two polygons

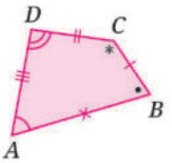
Two polygons are congruent if the following two conditions are satisfied together:

1. Their corresponding sides are equal in length.
2. Their corresponding angles are equal in measure.

If $ABCD$ and $EFGH$ are two polygons in which:



1. $AB = EF$, $BC = FG$,
   $CD = GH$, $AD = EH$

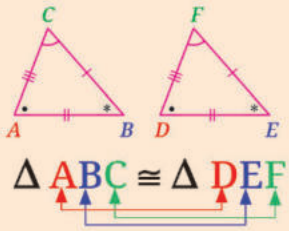


2. $m(\angle A) = m(\angle E)$, $m(\angle B) = m(\angle F)$,
   $m(\angle C) = m(\angle G)$, $m(\angle D) = m(\angle H)$

then The polygon $ABCD \cong$ the polygon $EFGH$

■ Note that

Congruent polygons must be named in the same order as their corresponding vertices. For example:



■ Remember that

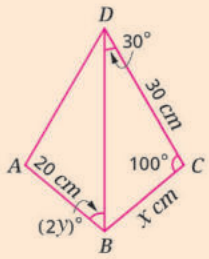
Reflection, translation, and rotation are geometric transformations that give an image congruent to the geometric shape because they preserve the following:

- Angle measures.
- Side lengths.
- Parallelism.

■ Self-Evaluation (1)

In the following figure:

ΔBCD is the image of ΔBAD by reflection in $\overleftrightarrow{DB}$

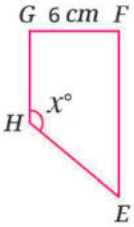


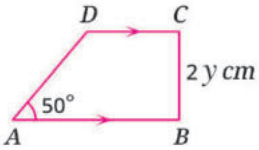
Find the value of each of x and y.

Example (1)

In the given figure:

If the polygon $EFGH$ is the image of the polygon $ABCD$ by rotation around a point, what is the value of each of x and y?





$\because \overline{AB} \parallel \overline{DC}$

$\therefore m(\angle A) + m(\angle D) = 180°$

(Two interior angles on the same side of the transversal $\overleftrightarrow{AD}$)

$\therefore m(\angle D) = 180° - 50° = 130°$

$\because$ The polygon $EFGH$ is the image of the polygon $ABCD$ by rotation

$\therefore$ The polygon $ABCD \cong$ the polygon $EFGH$

$\therefore m(\angle H) = m(\angle D) = 130°$ $\quad \therefore x = 130$

$BC = FG = 6$ $\quad \therefore 2y = 6$ $\quad \therefore y = 3$

Congruence of two triangles

The triangle is a polygon with 6 elements (3 sides and 3 angles).
To prove that two triangles are congruent, it is not necessary to prove that all corresponding elements are congruent. It is sufficient to prove that three elements, including at least one side, are congruent.